MILITARY SPECIFICATION
CIRCUIT CARD ASSEMBLY: 11745208
SCORING-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-48411 comprise the complete requirements for Scoring-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11745208	Circuit Card Assembly, Scoring-1

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item definition.- This assembly is one (1) of three (3) single ended printed circuit boards consisting of op-amps, flip-flops, mono-stable multivibrators, gates, diodes, and a digital to analog converter necessary to provide proper timing and analog signals for the digital readouts.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11745208.

3.2 Characteristics.

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3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power Supply.- The requirements of 3.2 shall be met with the following power supply voltages:

a. +2 VDC
b. +5 VDC
c. +10 VDC
d. +15 VDC
e. -15 VDC

3.2.1.2 Flip-Flops Type SN5474.- When a 4KHz clock pulse is applied to Pin 16, the complimentary outputs of the type SN5474 flip flops shall be as shown in Table I.

3.2.1.3 Flip Flops Type SN54L74.- When a 4KHz clock pulse is applied to Pin 16, the outputs of the type SN54L74 flip-flops shall be as shown in Table II.

3.2.1.4 Monostable Multivibrators.- When a 4KHz clock pulse is applied to Pin 16, the outputs of the monostable multivibrators shall be as shown in Table III.

3.2.1.5 4KHz Gated Clock Pulse Output.- With inputs applied as shown in Table III, output Pin 8 shall switch between Logic 0 and Logic 1 each time input Pin 16 is pulsed.

3.2.1.6 D/A Converter.- When the inputs are applied as described in paragraph 4.2.2.3.5, the D/A converter output Pin 35 shall range from 0 VDC to $\pm$ 10 VDC with .156 VDC resolution per input pulse and output Pins 5, 3 and 1 shall switch between Logic 0 and Logic 1.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11745208 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11745208 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

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3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and counter testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment. (See Figure 1)

4.2.1.1 Power Supply.- The power supplies shall be as follows:

a. +5.0 VDC ± 0.1 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

b. +10.0 VDC ± 0.1 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

c. +15.000 VDC ± 1 mVDC, 100 mA or greater and a ripple of less than 10 mV RMS.

d. +20.0 VDC ± 0.1 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

e. -20.0 VDC ± 0.1 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Pulse Generator.- The generator shall be capable of producing a single 10 nanosecond pulse, a single 10 nanosecond inverse pulse (5V 0V ⌐¯¯¯

4.2.1.5 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these test shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power Supply.- All requirements shall be met with the following power supplies:

a. +2 VDC ± 0.1 VDC
b. +5.0 VDC ± 0.1 VDC
c. +10.000 VDC ± 1 MVDC
d. +15.0 VDC ± 0.1 VDC
e. -15.0 VDC ± 0.1 VDC

4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

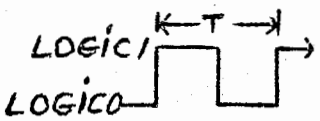
4.2.2.3 Test Set-Up.- The test of the circuit card asembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.3.1 Flip Flops Type SN5474 Test.- When the 4KHz clock pulse is applied to input Pin 16, the flip flop outputs shall be as shown in Table I. Output Pins 29 and 30 are inverses to each other, as are Pins 41 and 27.

TABLE I

Inputs

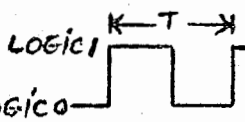
Pins 16



t = .25 msec ± .1 u sec
f = 4KHz ± 1Hz

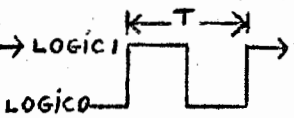
Outputs

Pin 29



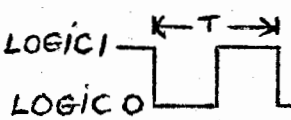
t = .5m sec ±.1u sec
f = 2KHz ± .5Hz

Pin 41



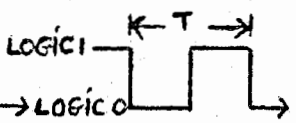
t = 1m sec±.2 u sec
f = 1KHz ± .25 Hz

Pin 30



t = .5 m sec ±.1 u sec
f = 2KHz ± .5Hz

Pin 27



t = 1m sec ± .2u sec
f = 1KHz ±.25 Hz

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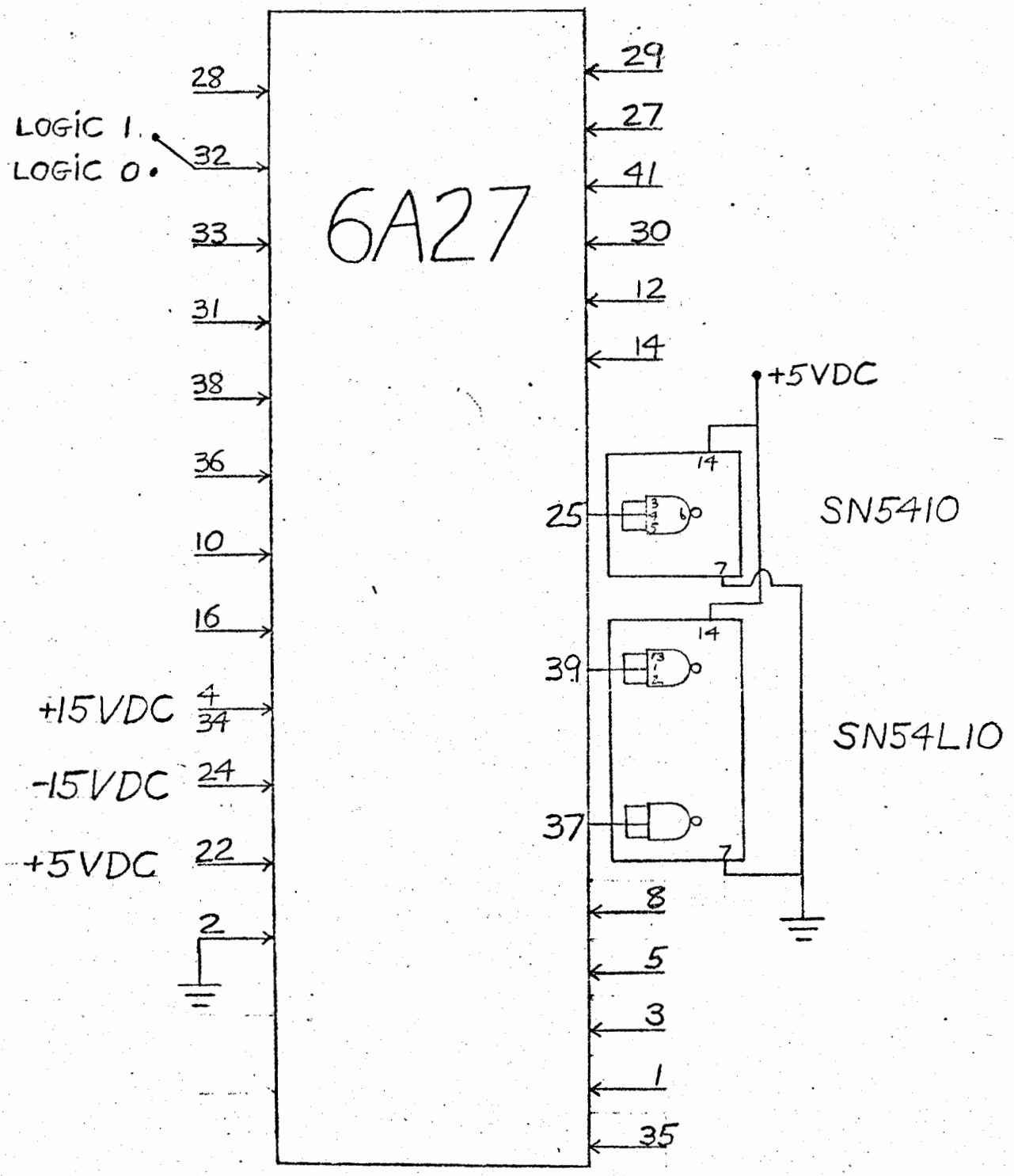
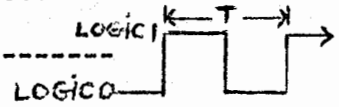


FIGURE 1

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4.2.2.3.2 Flip Flops Type SN54L74 Test.- Where the input instructions are followed, the flip flop outputs shall be as shown in Table II.

TABLE II

Input Instructions		Outputs	
Pin 32	Pin 16	Pin 39	Pin 37
Reset "0"	------	Logic 1	Logic 1
--------		Logic 0	Logic 0

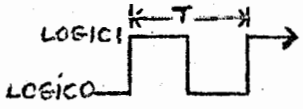
t = .25 m sec ±.1u sec
f = 4KHz ± 1Hz

4.2.2.3.2 Monostable Multivibrators Type SN54121 Test.- When a 4KHz clock pulse is applied to input Pin 16 the outputs of the monostable multivibrators shall be as shown in Table III. Output Pins 25 and 12 are the inverses of one another.

TABLE III

Input

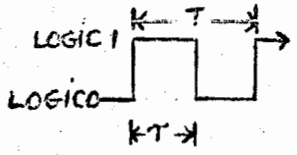
Pin 16



t = .25m sec ± .1u sec
f = 4KHz ± 1 Hz

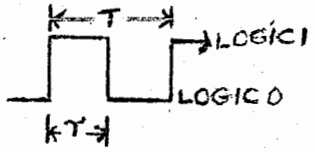
Output

Pin 14



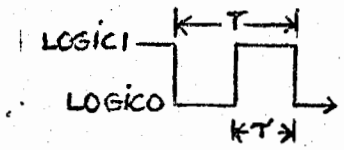
t = .25 m sec ±.1u sec
f = 4KHz ± 1Hz
= 70 u sec ±10 u sec

Pin 25



t = .25 m sec ± .1 u sec
f = 4KHz ±1Hz
= 7 u sec ± 1 u sec

PIN 12



t = .25 m sec ± .1 u sec
f = 4KHz ± 1Hz
= 7 u sec. ±1 u sec

4.2.2.3.4 Gated 4KHz Clock Test.- When the inputs are applied as stated, output Pin 18 shall function as shown in Table IV.

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TABLE IV

Inputs				Instruction	Output
Pin 10	Pin 28	Pin 38	Pin 36	Pin 16	Pin 18
Logic 1	Logic 1	0V ± 1mV	2V ± .1V	---- Apply several 10 nanosec. logic 1 pulses	----- Logic 1 ⇆ (1) Logic 0
Logic 1	Logic 1	2V ± .1V	0V ± 1mV	---- Apply several 10 nanosec. logic 1 pulses	----- Logic 1 ⇆ (1) Logic 0
Logic 0	Logic 1	2V ± .1V	0V ± 1mV	---- Apply several 10 nanosec. logic 1 pulses	----- Logic 0
Logic 1	Logic 0	2V ± .1V	0V ± 1mV	- - - - Apply several 10 nanosec. logic 1 pulses	- - - - Logic 1

Note: (1) Switches on each input pulse (Pin 16).

4.2.2.3.5 D/A Converter Test.- The D/A converter output voltage (Pin 35) shall range from 0 VDC to ± 10 VDC with .156 VDC resolution per input pulse when the inputs are applied as stated in the D/A converter test procedure below; also, output Pins 5, 3, and 1 will switch between Logic 0 and Logic 1.

D/A Converter Test Procedure

With a Logic 1 applied to Pin 10, 10V(1) ± 1 mV to Pin 38, 0 V ± 1 mV to Pin 36, Logic 1 (2) to Pin 38 and a Logic 1 (3) pulse to Pin 16 the output on Pin 8 shall be 0 V ± .2 V.

Read Pin 35 and apply a Logic 0 pulse to Pin 28, Pin 35 shall be +.156 V ± 10 mV greater than the previous voltage (4). For each additional Logic 0 pulse applied to Pin 28, Pin 35 shall increase +.156 V ± 10 mV per pulse until +10 V is reached at which point Pin 35 shall switch back to its initial value. This cycle will repeat with sufficient pulsing.

/to Pin 28

Apply 15 Logic 0 pulses, the output on Pin 5 shall switch from Logic 0 to Logic 1. Apply 15 additional Logic 0 pulses to Pin 28, on the 15th pulse the output on Pin 5 shall switch from Logic 1 to Logic 0.

Apply 2 Logic 1 pulses to Pin 16 and apply 30 Logic 0 pulses to Pin 28, the output on Pin 3 shall change from Logic 0 to Logic 1 on the 30th pulse. Apply 30 additional Logic 0 pulses to Pin 28, the output on Pin 3 shall change from Logic 1 to Logic 0 on the 30th pulse.

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Apply 2 Logic 1 pulses to Pin 16 and 2 Logic 0 pulses to Pin 28, the output on Pin 1 shall change from Logic 0 to Logic 1 on the 2nd pulse. Apply 2 additional Logic 0 pulses to Pin 28, the output on Pin 1 shall change from Logic 1 to Logic 0 on the 2nd pulse.

Note:

(1) All voltages are DC.
(2) With capability of switching to Logic 0 for 10 nansecond then back to Logic 1-referred to as Logic 0 pulses.
(3) All Logic 1 pulses are 10 nanosecond in duration.
(4) If Pin 35 voltage doesn't change, apply another Logic 1 pulse to Pin 16.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11745208.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1240-A617

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